

ADITYA BAYU PRATAMA

Lead Software & IT Engineer | Smart Manufacturing & Industry 4.0 Specialist

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PROFESSIONAL PROFILE

Results-driven **Lead Software & IT Engineer** with 7+ years of experience in Manufacturing Execution Systems (MES), Industrial IoT, Machine Vision, and Smart Factory Automation across high-precision electronics and oil & gas. Proven track record of scaling IT teams from 1 to 5 engineers, architecting custom C#/.NET enterprise solutions for 500+ product models, and executing digital transformations. **2-Time 1st Place Winner of Giken Global Value Engineering Award (2023 & 2025).**

CORE COMPETENCIES & TECHNICAL SKILLS

Languages & Frameworks: C# (.NET Framework), SQL (MySQL), Python, React.js, C++, LabVIEW, Dart/Flutter, REST APIs
Smart Manufacturing: MES Architecture, Industrial IoT (IIoT), Pokayoke Systems, AOI Machine Vision, Thermal Print Automation
Hardware & Engineering: PLC (BNSP Certified), Raspberry Pi, Arduino, Lidar Sensors, FMEA, DMAIC, Six Sigma Practices
IT & Leadership: Team Recruitment & Management (Led 5 Engineers), Network Infrastructure, Enterprise Server Maintenance

HONORS & GLOBAL AWARDS

 1st Place – Giken Global Value Engineering (2025) Top global prize for <i>Intelligent Machine Data</i> , optimizing real-time analytics & line efficiency.	 1st Place – Giken Global Value Engineering (2023) Won 1st place for <i>Manufacturing Traceability System</i> , eliminating defect passage across lines.
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PROFESSIONAL EXPERIENCE

GIKEN PRECISION INDONESIA <i>Lead Software & IT Engineer (Promoted from IT Engineer / Team Lead)</i>	Batam, Indonesia <i>Apr 2019 – Present</i>
Enterprise MES Architecture (C# & SQL): Designed and deployed a custom Windows MES managing 500+ product models across Automotive, Medical, and Electronics lines, enforcing Pokayoke defect prevention and CTQ quality compliance. Digital Transformation Platform (React.js / MySQL): Built an enterprise web application digitizing manual factory workflows, including employee records, org charts, overtime request approvals, ticketing tracking, and project scheduling. Industrial IoT Gate & ESD Safety Automation: Engineered an automated ESD wristband monitoring gate using Arduino, C#, and MySQL, enforcing 100% compliance before line entry and tracking hardware defect rates. Automated Thermal Printing Pipeline: Streamlined QR-code label printing workflows across electronic lines by building a direct C# and SQL interface to thermal printers, eliminating printing delays. Centralized Data Infrastructure: Established centralized network folder sharing servers across all departments, preventing hard drive failure data loss and boosting cross-departmental collaboration. IT Leadership & Department Growth: Recruited, mentored, and scaled the software and IT engineering department from a solo role to a 5-engineer team, managing NPI testing and digital infrastructure. Incident Response & System Diagnostics: Contained and eliminated a severe worm malware outbreak within 3 hours across 20 laptops without downtime; diagnosed core MySQL server faults to maintain 99.9% uptime; resolved factory network outages under strict 15-minute SLAs.	

CLADTEK BI-METAL MANUFACTURING <i>Junior Programmer – R&D Department</i>	Batam, Indonesia <i>Apr 2018 – Jan 2019</i>
Autonomous Welding Inspection Robot: Engineered a pipe-inspection robot using C#, Raspberry Pi, and Lidar sensors to autonomously navigate inside pipes, streaming real-time video for welding defect calculations. High-Speed Data Transmission: Developed low-latency protocols connecting embedded sensors to servers for real-time QA.	

SELECTED TECHNICAL & AI PROJECTS

- AI-Driven Grid Trading Application:** Developed a high-frequency trading bot in React.js and Telegram Bot API using zero-reasoning AI models and Binance API integration for ultra-low latency market execution.
- LabVIEW AOI PCB Inspection System:** Built an Automated Optical Inspection algorithm in LabVIEW Vision to detect missing or misaligned components on PCB lines, significantly reducing manual QA headcount and ST production time.
- IoT Emergency Response Finder Robot:** Built a Raspberry Pi/Arduino wireless rescue robot equipped with thermal, gas, PIR sensors, and live camera streaming controlled via web/mobile apps for hazardous search.
- Smart Multi-Biometric Access Controller:** Built an automated access control ecosystem using Python (OpenCV) for facial recognition, fingerprint validation, RFID, and PIN passcodes backed by MySQL logging.

EDUCATION & PROFESSIONAL CERTIFICATIONS

Bachelor of Applied Science (S.Tr.T.) - Mechatronics Engineering State Polytechnic of Batam <i>Graduated Cum Laude (GPA: 3.81 / 4.00)</i>	Key Certifications: - Google Data Analytics Professional • PLC Competency (BNSP) - Electrical Power Technician Class II (LPJK) • DevOps (Dicoding)
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